A DYNAMIC ONLINE WORKFLOW for TRANSLATING BIOMEDICAL ONTOLOGIES:

Showcasing the Translation of the Sickle Cell Disease Ontology (SCDO)

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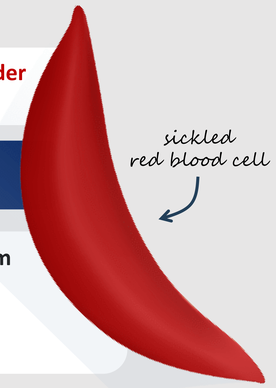
BACKGROUND & MOTIVATION

SCD & SCDO:

Sickle Cell Disease (SCD) is a **complex multisystem red blood cell disorder** and one of the world's most common monogenic pathologies.

SCD has its **highest incidence and prevalence in Sub-Saharan African countries**, where **>70% of sufferers live** [1].



The **Sickle Cell Disease Ontology (SCDO)**[2] **supports the SickleInAfrica Consortium in harmonising SCD data from multiple, mostly African, countries.**

→ Many of these countries are **non-English-speaking**,

→ with **all 21 Francophone African countries being affected by SCD.**

Previous Translation Method:

Previously developed a **novel method for developing and managing multilingual ontologies** [3]:

→ To **ease harmonisation** of SCD data from non-English speaking countries.

→ To make SCDO-driven **health-tracking, diagnostics and therapeutics applications more accessible** to non-English speaking patients and clinicians.

The translation method **included using**:

→ an **Ontology Translation Tool (OTT)** that we developed (Figure 1).

→ a **language tag protocol** that we developed.

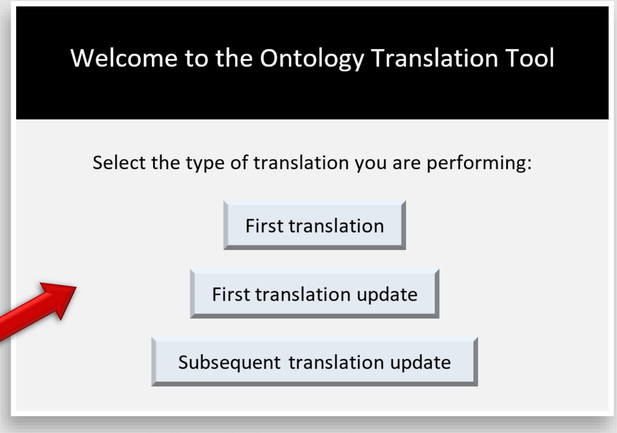


Figure 1. OTT's first window seen by users.

Reasons for Updating Translation Method:

To use **Babelon format TSV files** to:

→ reuse translations from other ontologies

→ easily share SCDO translations

For a faster and more robust method of managing multiple language translations at varying degrees of completion

To **produce relevant files** for:

→ viewing translations on publicly available ontology repositories

→ easily updating SCDO website content

METHODS & RESULTS

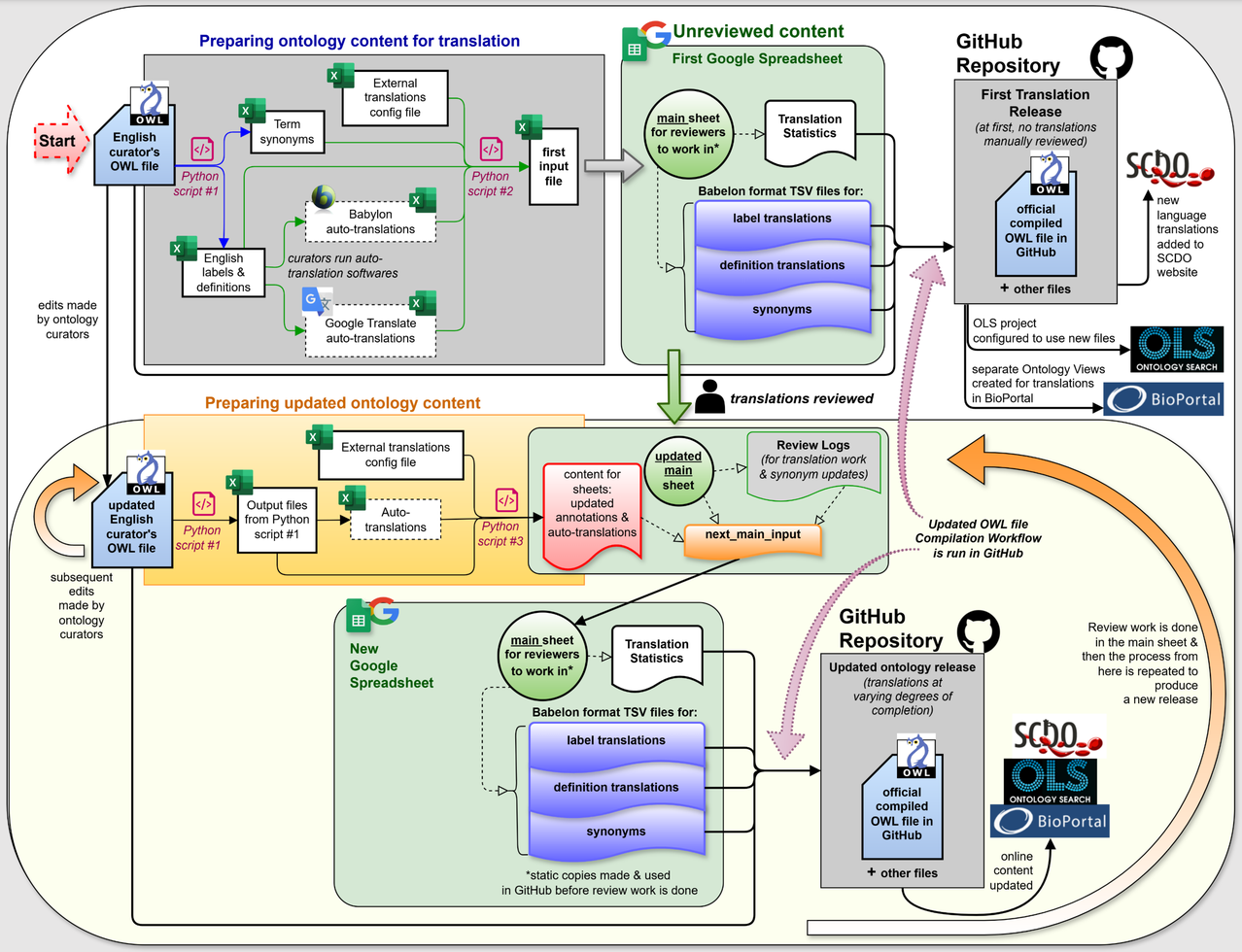


Figure 2. Updated Workflow for Dynamic Online Ontology Translation Creation and Management.

**Once complete, the workflow will be made available via the SCDO GitHub repository.*

RELEVANCE

An easily reusable **dynamic online solution for managing ontology translation projects** that:

→ allows for:

- **shorter turn-around times** for ontology releases
- **easy management** of multiple language translations at different stages of completion
- **seamless reuse and attribution** of translations produced by other biomedical ontologies

→ produces:

- **files** for:
  - » **viewing translations** on main biomedical **ontology repositories**
  - » **SCDO translations to be reused** by other ontologies
- **auto-generated statistics** for each release

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1. Thomson, A. M., McHugh, T. A., Oron, A. P., et al. (2023). Global, regional, and national prevalence and mortality burden of sickle cell disease, 2000-2021: a systematic analysis from the Global Burden of Disease Study 2021. Lancet Haematol. 10(8), e585 - e599. doi:10.1016/S2352-3026(23)00118-7
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